# BAYER SALES INTELLIGENCE

Whole-System Build Plan

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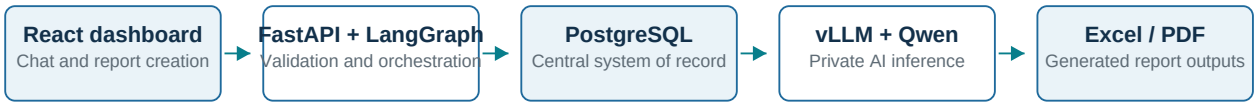
| **Purpose**<br>A secure, AI-powered sales intelligence platform for daily data imports, precise business questions, trend analysis, charts, and report creation. | **Delivery model**<br>Private AWS deployment with a centralized PostgreSQL database, company SSO, and private vLLM/Qwen inference. |

## Target outcome

- Daily Excel sales reports are validated, deduplicated, audited, and imported into one centralized PostgreSQL database.
- Internal users ask precise sales questions in natural language and receive evidence-backed answers, tables, trends, and charts.
- The Create Report dashboard captures all report choices before generation. Excel is the primary report format; PDF is optional.
- No vector database, embeddings, or RAG layer is included. PostgreSQL remains the single source of truth.

## Architecture at a glance




*FastAPI is the sole database-access layer. The AI model and sandbox never receive database credentials.*

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| **Central database**<br>Amazon RDS PostgreSQL stores sales, imports, business metadata, audit logs, report history, and workflow state. | **Private AI compute**<br>A GPU EC2 instance runs vLLM and Qwen behind an internal API. It receives only approved results, not database access. | **Secure operations**<br>Company SSO, private networking, encrypted storage, least-privilege roles, backups, logs, and monitored services. |

# 1. Platform and service architecture

AWS provides the private network, centralized database, durable file storage, and managed identity boundary. Application services are deployed privately and users access the dashboard over authenticated HTTPS.

| Layer | Technology / location | Responsibility |
|---|---|---|
| Web application | React on S3 + CloudFront | Chat workspace, import administration, report builder, result tables, charts, and downloads. |
| Identity | Company SSO via Amazon Cognito federation | Authenticate users and issue JWTs. Administrator and analyst roles are enforced by FastAPI. |
| Application API | FastAPI in private containers | Only service permitted to access PostgreSQL. Owns validation, queries, imports, reports, and response streaming. |
| Workflow engine | LangGraph inside FastAPI | Routes each request through deterministic, auditable steps and persists workflow state for retries. |
| AI inference | Qwen on private GPU EC2 via vLLM | Interprets requests and writes explanations via an internal OpenAI-compatible API. No database credentials. |
| Data | Amazon RDS PostgreSQL | Central system of record for sales data, metadata, audit history, reports, and workflows. |
| Files | Private S3 buckets | Stores uploaded source files, generated Excel/PDF reports, and controlled download references. |
| Custom calculations | Private isolated worker | Runs generated Python only in short-lived, restricted containers; returns a calculation result and execution log. |

# 2. Data ingestion and analytics

- Define a canonical sales model with a row-level sales fact table, date, route, product/SKU, customer, sales representative, coffee-machine, and other business dimensions supplied during discovery.
- Process each daily Excel file through upload, staging, column mapping, type and required-field checks, duplicate detection, business-rule validation, transactional import, and audit recording.
- Maintain a PostgreSQL business catalog for approved measures, filters, aliases, fiscal-calendar logic, business definitions, and permitted comparison periods.
- Compile model requests into parameterized, read-only SQL. Do not allow the model to create or execute raw SQL directly.
- Implement common measures as tested Python functions: totals, growth, percentage change, rankings, contribution, variance, moving averages, trend detection, and period/season exclusions.

# 3. AI workflow, safety, and user experience

LangGraph executes a controlled workflow rather than allowing an autonomous agent to freely access systems.

| Step | Workflow behavior |
|---|---|
| 1. Interpret | Qwen converts the question into a typed request: measures, date range, filters, grouping, comparison, chart need, and report intent. |
| 2. Validate | FastAPI resolves aliases and checks every requested field against the approved business catalog. Ambiguity produces a clarification request. |
| 3. Retrieve | FastAPI compiles and runs parameterized read-only SQL under a restricted database role, returning only required rows or aggregates. |
| 4. Calculate | Route common analytics to the tested library. Send only genuinely novel calculations to the isolated Python worker. |
| 5. Visualize | Create a validated chart specification: time series for trends, bar charts for comparisons, and composition charts for shares. |
| 6. Explain | Provide the model with approved results, filters, and definitions. It writes the English answer and observations grounded in that evidence. |
| 7. Report | When requested, use the same report specification and data to create an Excel workbook and optional PDF. |

## Security controls

- vLLM has no PostgreSQL credentials. The sandbox has neither database credentials nor general network access.
- Generated code is AST-validated and runs in a short-lived container with read-only approved input, CPU/memory/time limits, and execution logs.
- Encrypt data at rest and in transit; store secrets in AWS Secrets Manager; restrict database and service access through private security groups and least-privilege IAM roles.
- Record import activity, executed analytics request, query provenance, report generation, sandbox runs, and security-relevant failures for auditability.

# 4. Reporting and API surface

The Create Report dashboard is the report-configuration surface; generated reports are final, downloadable outputs and are not edited after creation.

- Start with title, date range, filters, measures, grouping, comparisons, insight/recommendation inclusion, chart selection, and output format. Keep the form schema-driven for the final fields provided during discovery.
- Generate Excel as the primary format with a summary sheet, selected analysis/data sheets, a source-and-filters sheet, and embedded selected charts.
- Generate PDF only when selected, based on exactly the same specification and figures as Excel.
- Expose authenticated endpoints for imports and import status, catalog access, chat messages and streamed responses, report creation/status, and report downloads.

# 5. Delivery sequence

| Phase | Major deliverable |
|---|---|
| 1. Discovery and data contract | Representative Excel files, row grain, column mapping, business definitions, duplicate rules, aliases, fiscal calendar, source ownership, and final report-builder fields. |
| 2. Cloud foundation | VPC, private networking, RDS, S3, Cognito/SSO federation, secrets, container registry, CI/CD, encrypted backups, monitoring, and logging. |
| 3. Data foundation | Schema migrations, staged import pipeline, validation/deduplication, audit trail, administrator import UI, and verified sample-data imports. |
| 4. Analytics foundation | Business catalog, query compiler, prebuilt calculation library, chart contract, and known-answer automated tests. |
| 5. AI workflow | Private vLLM/Qwen deployment, LangGraph nodes, streamed chat UI, evidence/provenance display, safety guardrails, and evaluation set. |
| 6. Reporting | Create Report dashboard, Excel templates and embedded charts, optional PDF generation, file storage, history, and downloads. |
| 7. Advanced calculations | Sandbox worker, generated-code validation, execution records, failure handling, and understandable custom-calculation output. |
| 8. Production readiness | Load/security testing, backup/restore drill, model-answer evaluation, user acceptance testing, runbooks, monitored launch, and post-launch tuning. |

# 6. Acceptance criteria

- A valid daily Excel file imports once; invalid rows are reviewable; repeated source files and sales rows cannot produce duplicate facts.
- Supported questions return validated, parameterized analytics results matching independently verified expected outputs.
- The AI cannot access database credentials, write to PostgreSQL, execute arbitrary SQL, bypass roles, or state figures absent from returned evidence.
- Unsafe or over-limit sandbox code is blocked without affecting production systems or data.
- Excel reports include selected filters, periods, tables, calculations, and charts; optional PDFs contain the same figures.
- Central database recovery is proven through a restore drill; failures in application or model services do not corrupt data or erase audit history.

## Implementation assumptions

- AWS is the selected cloud provider. PostgreSQL is centrally hosted in Amazon RDS from the first release.
- Private AWS GPU hosting is the initial AI-compute approach; final model quantization and GPU size are chosen after a controlled performance and answer-quality benchmark.
- Qdrant, embeddings, vector search, and RAG are excluded. Excel is the primary report format and PDF is optional.